MODELOFF™

Financial Modeling World Championships 2015

Section 3: Case Study – All About Performance

Relates to Questions 17 – 26

35 Marks available in this Section - Estimated time is 35 minutes

INTRODUCTION

Your firm, Health Services Limited, is bidding on a contract to provide ambulance services to a city for the five calendar years from 2016 to 2020.

The contract is paid and measured quarterly. You will be paid $7m per quarter to operate this contract and this will cover up to 20,000 responses to calls. For each additional 100 calls (or part thereof) per quarter, there will be an extra payment made of $18,000. Each call is given a priority rating of Emergency, Urgent or Normal.

The contract is subject to certain performance standards. Deductions will be made from the payment where these standards are not met. You must carry out the contract for the full five years. You cannot abandon the contract if the quarterly cash flow become negative.

You are required to analyse the financial impact of this contract based on some assumptions about expected call volumes and performance. Your firm requires all analysis to be done on a real basis using a real, pre-tax discount rate of 10% per annum where necessary. All discounting should be done using 31 December 2015 as the base date. Assume all cash flows are paid/received at the end of each quarter. When calculating the periodic discount factors for present value questions, treat each quarter as being exactly 0.25 years long.

PERFORMANCE DEDUCTIONS

The contract is subject to performance terms that dictate the response time to each type of priority rating. The response time to each call is measured in minutes. Deductions in any quarter are capped at the total pre-deduction payments for that quarter.

Emergency calls

- Response time should be no more than 7 minutes for all calls.
- Deduction per call of $500 for every minute or part thereof that the call's response time is over the 7 minute response requirement.
- The deduction for emergency calls is capped at $2,000 per call

Urgent calls

- Response time should be no more than 12 minutes for 95% of calls.

- Z is defined each quarter as the percentage of urgent calls with a response time of no more than 12 minutes.
- Where Z is less than 95% for a quarter, the deduction for that quarter is calculated as:
  $3,000 x (95% - Z%) x (total urgent calls).

Normal calls

- Deduction is calculated against the average response time for all normal calls in a quarter.
- The Average Response Time for normal calls is referred to as 'ART'.
- The deduction each quarter is calculated under one of the following bands:
  - **Band 0** If ART is less than 18 minutes, no deduction
  - **Band 1** If ART is greater than or equal to 18 minutes and less than 20 minutes, the deduction is calculated as ($100 x # normal calls)
  - **Band 2** If ART is greater than or equal to 20 minutes and less than 22 minutes, the deduction is calculated as ($110 x # normal calls)
  - **Band 3** If ART is greater than or equal to 22 minutes and less than 24 minutes, the deduction is calculated as ($130 x # normal calls)
  - **Band 4** If ART is greater than or equal to 24 minutes, the deduction is calculated as ($150 x # normal calls)

ASSUMPTIONS

Expected number of calls:

- **Emergency** 5,000 in the first quarter, growing at 1.25% per quarter thereafter
- **Urgent** 6,000 in the first quarter, growing at 1% per quarter thereafter
- **Normal** 8,000 in the first quarter, growing at 0.5% per quarter thereafter
- Do not round the number of calls when calculating the number of calls or when calculating deductions.

Expected costs

- Ambulances are leased at a cost of $80,000 per ambulance per quarter. There are 24 leased in the first quarter.
- Staff are employed at a cost of $25,000 per staff member per quarter. There are 96 staff employed in the first quarter.
- The number of ambulances and staff employed should grow at the same percentage as the total number of calls. Do not round the number of ambulances or staff.
- Other costs of $300,000 per quarter

Expected performance

- **Emergency** 95% of emergency calls have a response time within 7 minutes, 3% within 8 minutes, 1% within 9 minutes, 0.75% within 10 minutes and 0.25% within 11 minutes (do not round the number of calls).
- **Urgent** 97% of urgent calls are reached within the 12 minutes in the first quarter. The percentage falls by 0.25 percentage points per quarter.
- **Normal** The Average Response Time in the first quarter is 16 minutes. This increases by 2.0% per quarter.

There is no workbook provided for this section. You will need to create your own.

QUESTIONS

Question 17

What is the total number of calls made in the third quarter of 2019? [3 marks]

a. 21,420 calls

b. 21,421 calls

c. 21,422 calls

d. 21,423 calls

e. 21,424 calls

f. 21,425 calls

Question 18

What is the total revenue (excluding performance deductions) earned in the first quarter of 2020? [3 marks]

a. $7,321,000

b. $7,322,000

c. $7,323,000

d. $7,324,000

e. $7,325,000

f. $7,326,000

Question 19

What is the total cost (excluding any performance deductions) for the 2018 year? [3 marks]

a. $19,944,613

b. $19,944,614

c. $19,944,615

d. $19,944,616

e. $19,944,617

f. $19,944,618

Question 20

What is the performance deduction due to emergency calls in the first quarter of 2016? [4 marks]

a. $205,750

b. $206,000

c. $206,250

d. $206,500

e. $206,750

f. $207,000

Question 21

What is the performance deduction due to urgent calls in the fourth quarter of 2019? [4 marks]

a. $365,701

b. $365,702

c. $365,703

d. $365,704

e. $365,705

f. $365,706

Question 22

What is the performance deduction due to normal calls over the whole contract? [4 marks]

a. $13,139,465

b. $13,139,466

c. $13,139,467

d. $13,139,468

e. $13,139,469

f. $13,139,470

Question 23

What is the net present value of the contract? [4 marks]

a. $18,707,961

b. $18,707,962

c. $18,707,963

d. $18,707,964

e. $18,707,965

f. $18,707,966

Question 24

You are considering a new preventative healthcare initiative that will mean no growth in total call volumes over the five years. This will involve an additional fixed quarterly cost of $100,000 to achieve. What is the net present financial benefit/cost of this initiative? Do not round the $ values in your model. [5 marks]

a. $2,996,973 benefit

b. $2,996,974 benefit

c. $2,996,975 benefit

d. $2,996,976 benefit

e. $2,996,977 benefit

f. $2,996,978 benefit

Question 25

Reset any one-off changes you may have made for the previous question before attempting this question.

You estimate you can reduce the total cost of deductions each quarter by 50% if you increase the initial number of ambulances by 25% and the initial number of staff by 25%. What is the net present financial benefit/cost of this initiative? Do not round the $ values in your model. [5 marks]

a. $10,604,524 cost

b. $10,604,523 cost

c. $10,604,522 cost

d. $10,604,521 cost

e. $10,604,520 cost

f. $10,604,519 cost

Question 26

Please upload your workbook for this section. [0 marks]